



Siddhi Vaity

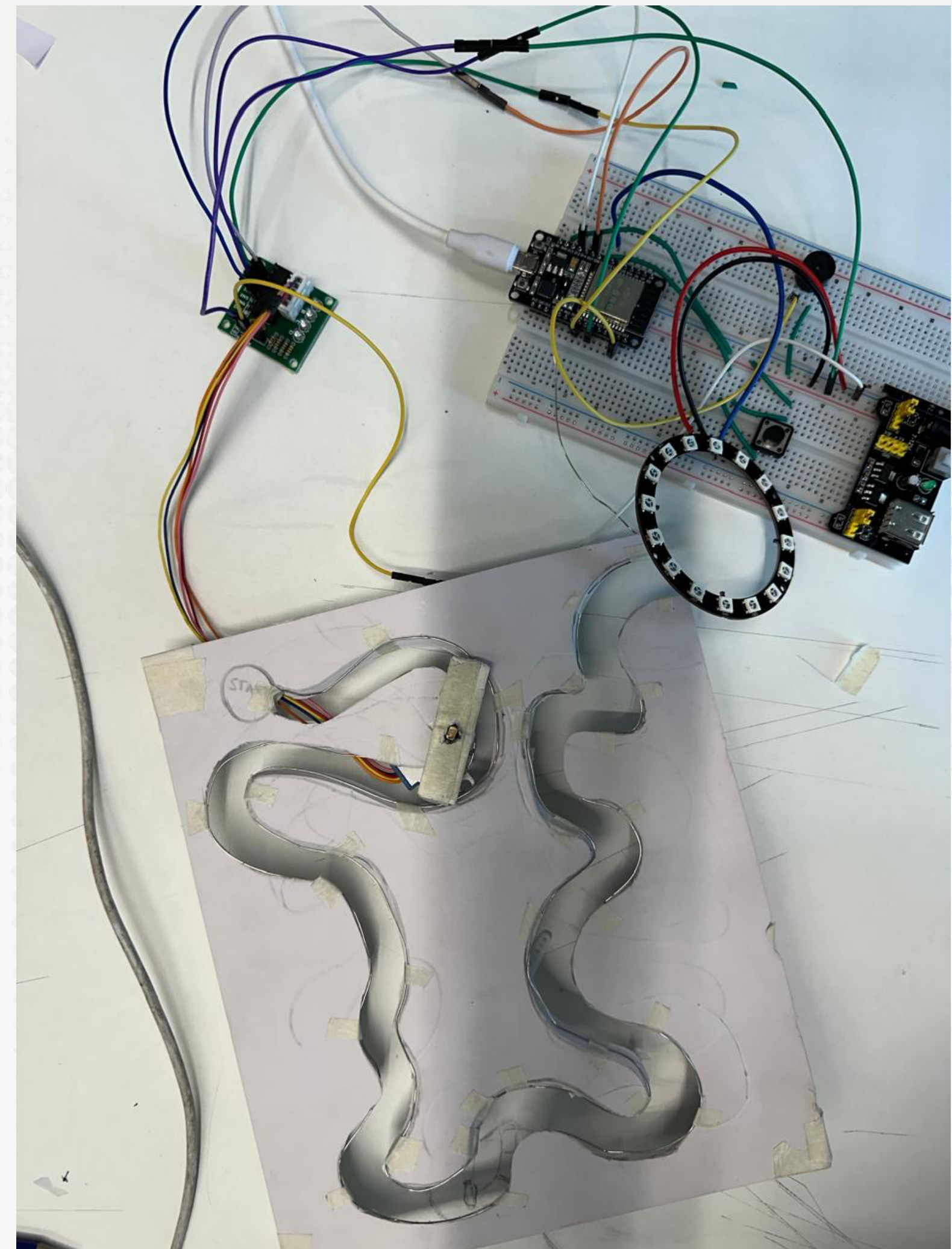
Anindita Vaishnava Hazra

Steady Hands

How to Play ?

Be careful! Do not touch the edges.
If the buzzer goes off,
then GAME OVER

The player must move carefully along a path
without making contact, and the moment skin
touches the conductive surface, the system
responds.



WHAT led us to this idea?

we chose to create Steady Hands as our assessment project for the Open Design and Technology module because we wanted to explore the concept of capacitive touch.

We were interested in how capacitive touch allows interaction through the human body itself, without using buttons or traditional switches.

The concept of a “steady hands” game felt appropriate because it translates a simple electronic principle into a physical challenge.

This game makes capacitive touch visible and understandable through play. Rather than explaining the concept theoretically, the game demonstrates it through direct feedback.

HOW we decided to use specific components?

We first determined technical feasibility by evaluating whether the concept could be reliably executed using accessible components and simple logic, and learning whether :

The ESP32 supports touch sensing, digital input/output, and NeoPixel control.

It can detect contact and trigger immediate feedback through a buzzer and LEDs.

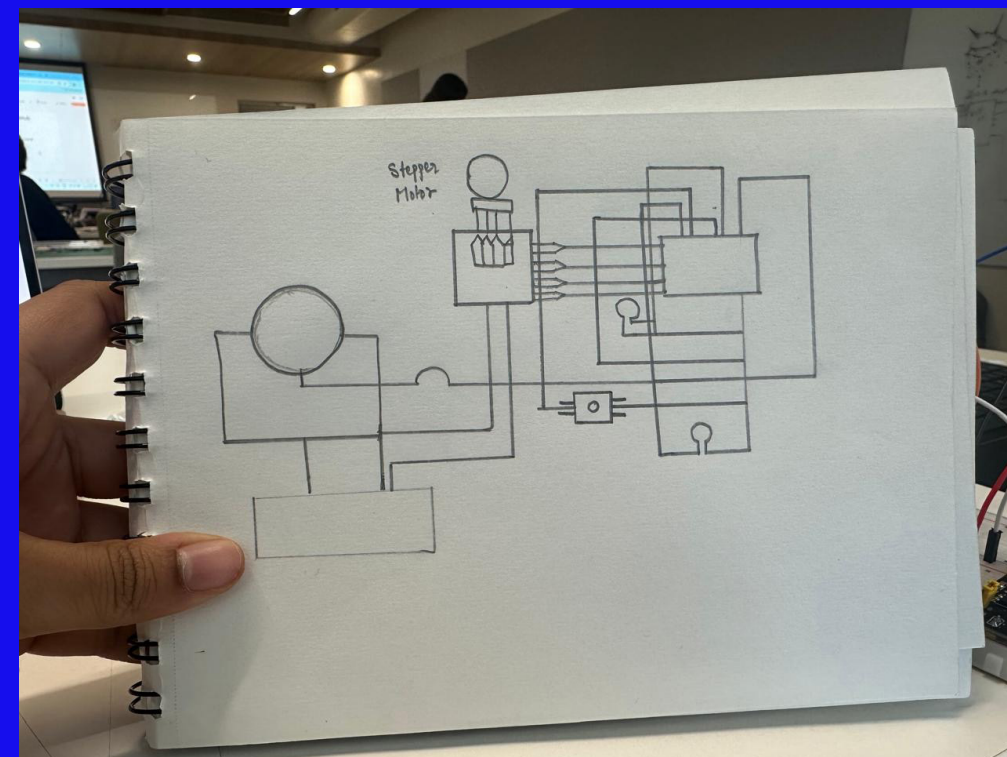
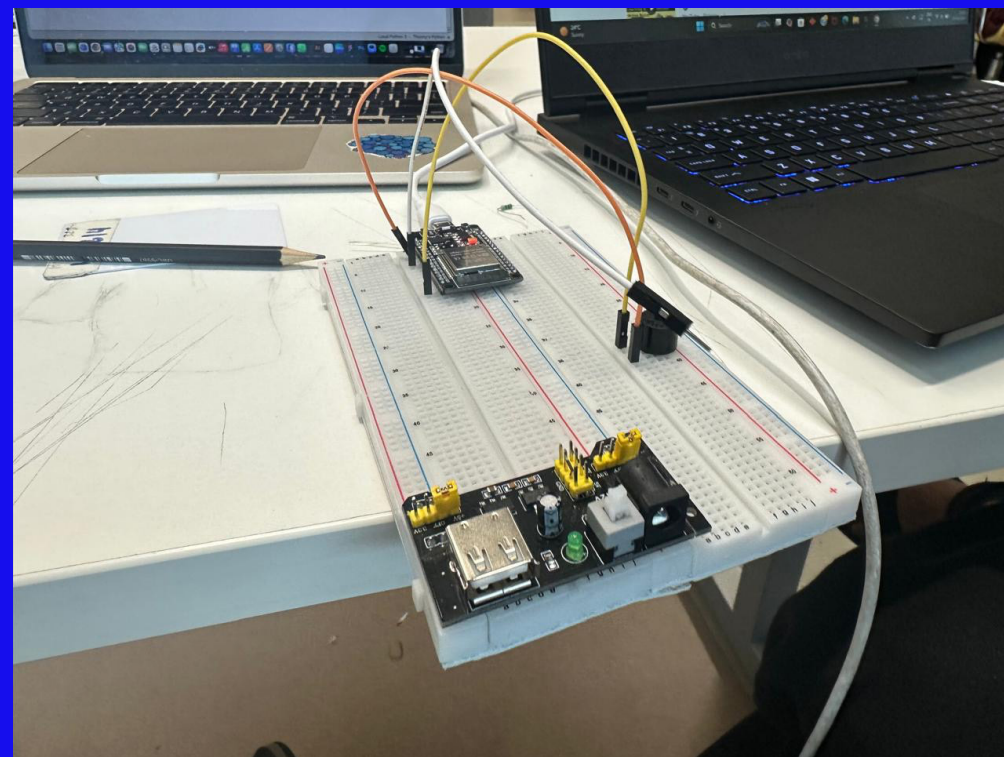
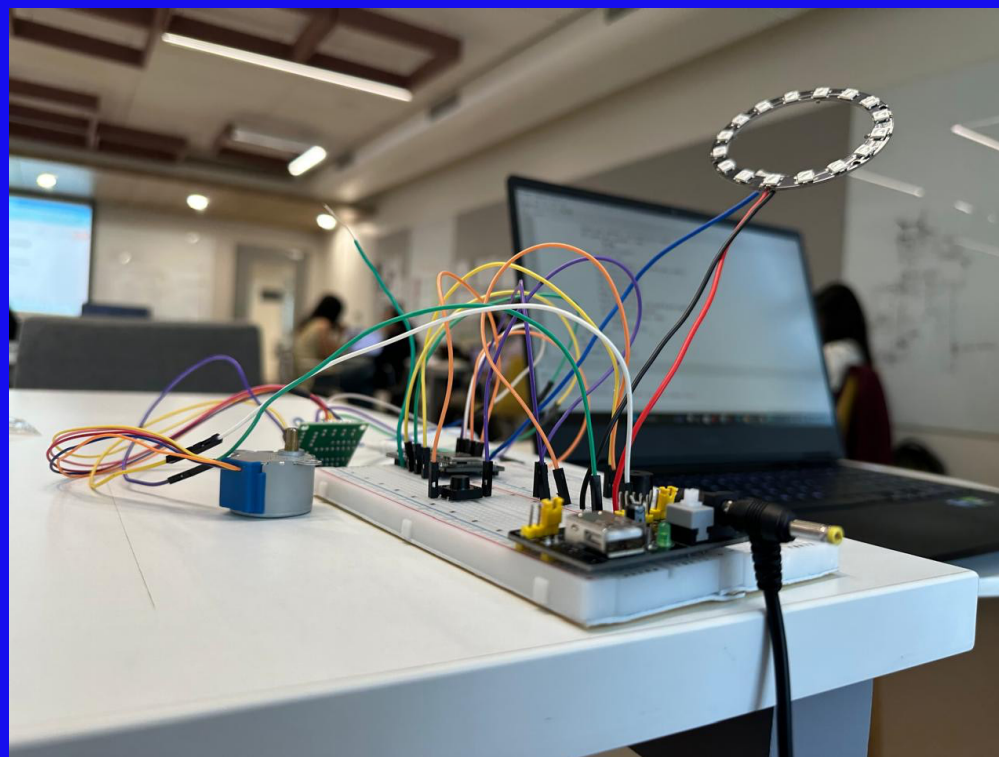
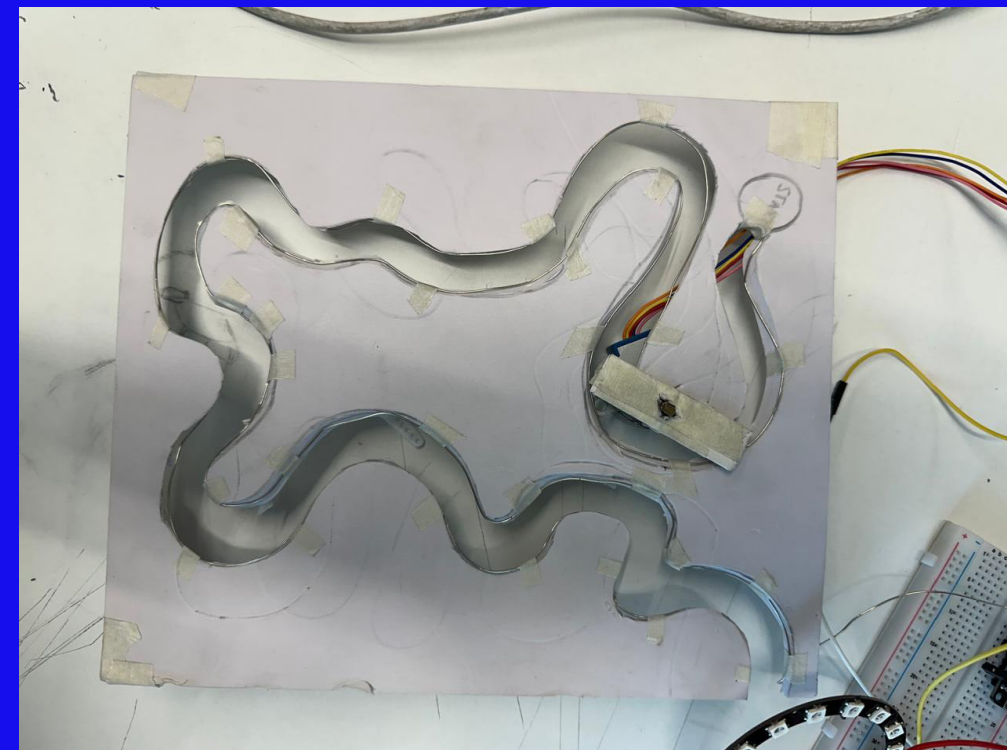
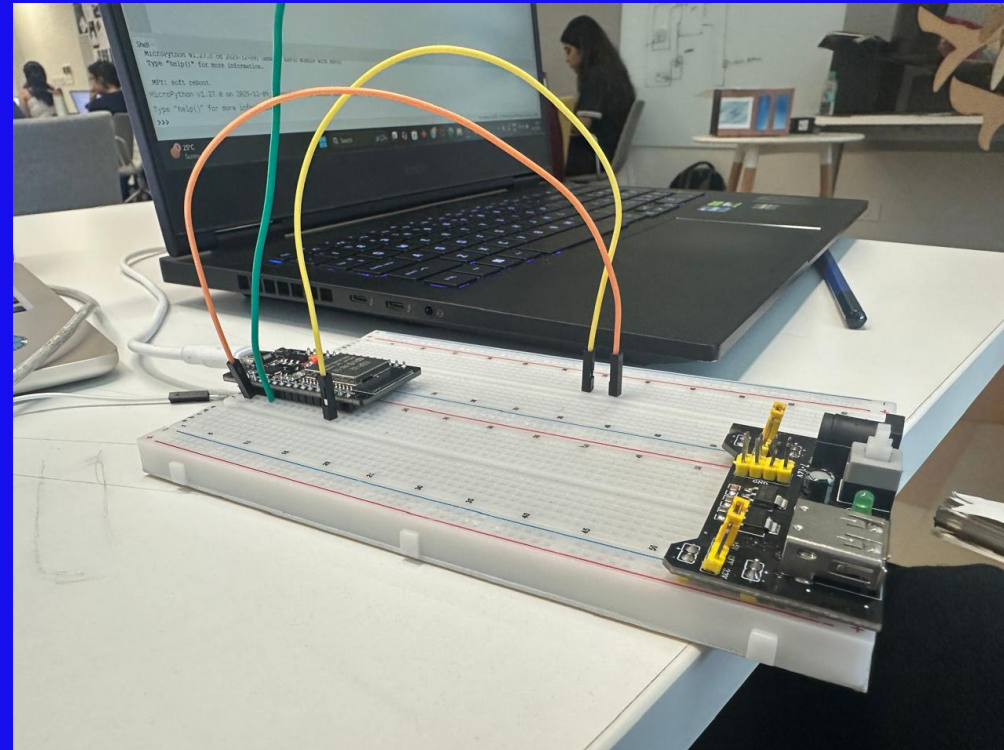
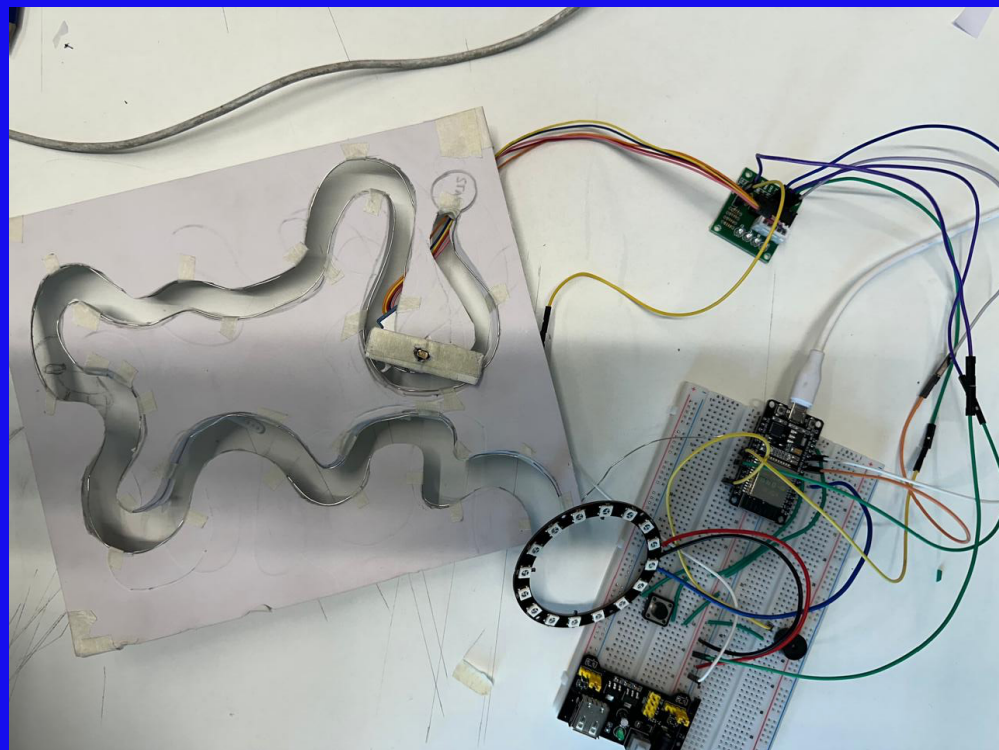
The wiring mechanism, using copper edges as a contact trigger, is straightforward and low-risk to implement.

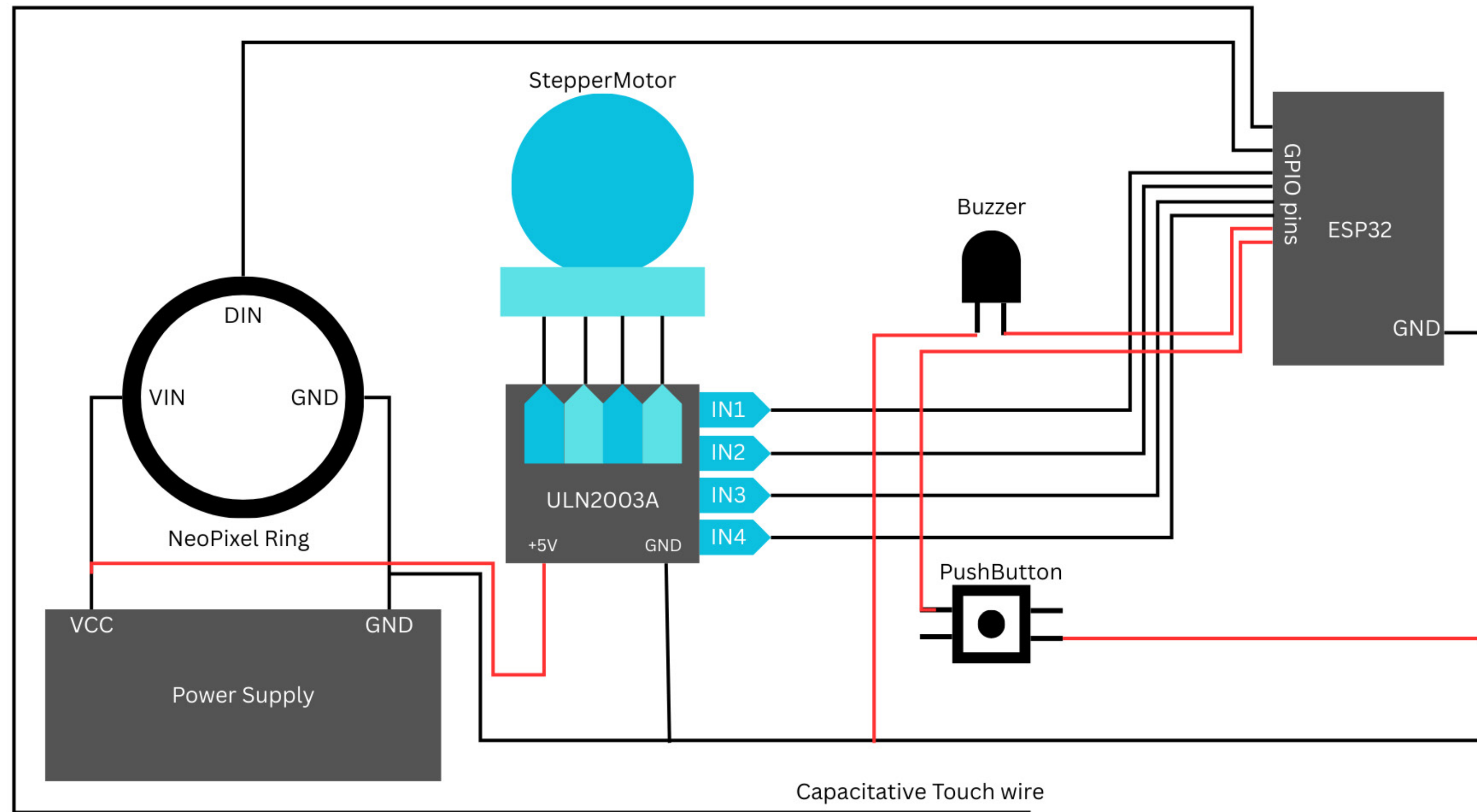
The system relies on basic loops and conditional logic, keeping programming complexity manageable.

The idea aligns well with the hardware capabilities.

Overall, the project is practical, buildable, and responsive in real time.

Documentation





Circuit Diagram

Code

```
from machine import Pin, TouchPad, PWM
import time
import neopixel

my_touch_pin = TouchPad(Pin(4))

buzzer1 = Pin(22, Pin.OUT)

my_neo = neopixel.NeoPixel(Pin(33), 16)

pb = Pin(25, Pin.IN, Pin.PULL_UP)
in1 = Pin(27, Pin.OUT)
in2 = Pin(5, Pin.OUT)
in3 = Pin(19, Pin.OUT)
in4 = Pin(12, Pin.OUT)
seq = [[1,0,0,0],[0,1,0,0],[0,0,1,0],[0,0,0,1]]

count = 0
```

```
while True:

    touch_pin_values = my_touch_pin.read()
    if touch_pin_values < 300:
        buzzer1.value(1)
        print(touch_pin_values)
        time.sleep(0.2)
    else:
        buzzer1.value(0)
        print(touch_pin_values)
        if touch_pin_values > 300:
            for _ in range(2000):
                touch_pin_values = my_touch_pin.read()
                if touch_pin_values < 300:
                    break
                print("heeh")
            for step in seq:
                touch_pin_values = my_touch_pin.read()
                in1.value(step[0])
                in2.value(step[1])
                in3.value(step[2])
```

```

in4.value(step[3])
time.sleep(0.005)
touch_pin_values = my_touch_pin.read()
print(touch_pin_values)
if touch_pin_values < 300:
    break

val = pb.value()
if val == 0:
    while True:
        for i in range (0,16):
            my_neo[i] = (0, 0, 255)
            my_neo.write()
            time.sleep(0.1)
        for i in range (0,16):
            my_neo[i] = (0,0,0)
            time.sleep(0.1)
            my_neo.write()

```

The logic was straightforward yet complex, we found it challenging to put different loops under an existing loop, putting a "print" statement helped in that case.

Using multiple break statements so that the microprocessor reads the values everytime in the loop so that it breaks the loop and detects touch instantaneously ,

One of the key learnings were changing the Capacitive Touch Value , so that the wire in contact with the surface of the cardboard does not detect the touch of the cardboard, and only detects touch of the human finger.

Individual contribution

Siddhi:

- The final code
- Brainstorming the circuit diagram
- Making the electric circuit on board
- Presentation
- Final Digital Circuit Diagram

Anindita:

- Final circuit diagram
 - initial ideation
- physical model of our game (cardboard, hardboard)
 - Presentation
 - Documentation of work

Link to the unlisted video on
youtube for our project:

<https://youtu.be/TzbNoGfEmKI>